

PHYS6006 Synoptic coursework: Exam-style questions

Apurba Biswas - ID 27894649

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1 A question: 5 marks

Given the time-dependent Schrodinger equation with a general spatially varying potential $U(x)$, derive the time-independent Schrodinger equation $\hat{H}\psi(x) = E\psi(x)$. Define the Hamiltonian $\hat{H}$ and the energy E

$$i\hbar \frac{\partial \Psi(x, t)}{\partial t} = \frac{\hbar}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} + U(x)\Psi(x, t)$$

2 A question : solution

Presume the solution is seperable i.e. $\Psi(x, t) = \psi(x)T(t)$

(1 mark)

$$-\frac{\hbar}{2m} \frac{d^2 \psi(x)}{dx^2} T(t) + U(x)\psi(x)T(t) = i\hbar \frac{dT(t)}{dt} \psi(x)$$

Diving by $\psi(x)T(t)$ leads to

$$-\frac{\hbar}{2m} \frac{1}{\psi(x)} \frac{d^2 \psi(x)}{dx^2} + U(x) = i\hbar \frac{1}{T(t)} \frac{dT(t)}{dt}$$

(1 mark)

Both sides are independent (i.e. x and t terms have been separated) and thus both sides must be independently constant

(1 mark)

Equation the RHS to a constant E , and solving for $T(t) = e^{-iEt/\hbar}$ (1 mark)

$$-\frac{\hbar}{2m} \frac{1}{\psi(x)} \frac{d^2 \psi(x)}{dx^2} + U(x) = E\psi(x)$$

$$\hat{H}\psi(x) = E\psi(x) \quad \hat{H} = -\frac{\hbar}{2m} \frac{d^2}{dx^2} + U(x)$$

(1 mark)

3 B question : 20 marks

This question about Blackbody radiation. We derive the Plank's blackbody distribution, considering the classical and quantum approaches. This question uses knowledge from many modules including Waves, Light and Quanta, Quantum Physics of Matter, Energy and Matter, Waves, and Quantum Physics.

This question maybe considered somewhat lengthy and difficult. However, I think this question is sensible as we've performed the calculation several times. Also, most of the questions give the expressions to be verified, so the student can still attempt majority of the question, even if they have been unable to complete the previous parts. I think a well prepared physics student would be able to achieve full marks in the question. This question is essentially all recall as all of the content has been presented in lectures and notes.

Consider an electromagnetic wave in a 1 dimensional box of length L . Let the wavefunction of the wave be $\psi = A \sin(kx) + B \cos(kx)$

1. Given the boundary conditions $\psi(0) = \psi(L) = 0$, deduce:

- the constants A and B
- the allowed wavevectors k_n for a particular mode n

(4 marks)

2. Write down the allowed wavevectors k_x, k_y, k_z for an electromagnetic wave in a 3 dimensional cube with sides length L and give an expression for the angular frequency ω for such a wave

(1 mark)

3. Given that each allowed frequency ν occupies a volume $(c/2L)^3$ in the frequency space, derive the number of modes with frequencies in range ν to $\nu + d\nu$. Verify it is equal to:

$$N(\nu) = \frac{8\pi L^3}{c^3} \nu^2 d\nu$$

(Hint : Assume the size of the box L to be large and consider the frequencies)

(2 marks)

4. From Maxwell's equation, recall that each mode has a kinetic and potential term and hence 2 degrees of freedom. Using the equipartition theorem, prove that the expression for the average thermal energy per unit volume in the frequency range ν to $\nu + d\nu$ is given by Rayleigh-Jeans law:

$$u(\nu) d\nu = \frac{8\pi k_B T}{c^3} \nu^2 d\nu$$

(2 marks)

5. Using the expression, explain above how this classical approach leads to the so called ultraviolet catastrophe?

(1 mark)

6. To circumvent the ultraviolet catastrophe, we follow Planck's argument instead. Planck considered a single mode with frequency ν and assumed that the radiation at this frequency can only occur in chunks of energy $\epsilon = h\nu$ where h is Planck's constant and ϵ is the energy.

Given that the probability of having m such quanta is proportional to the Boltzmann distribution i.e. $P(\epsilon) \propto e^{-\frac{m\epsilon}{k_B T}}$, by firstly normalising the distribution, show that the average energy in this mode is given by:

$$\frac{h\nu}{e^{h\nu/k_B T} - 1}$$

and hence derive the Planck's Blackbody distribution

$$u(\nu) d\nu = \frac{8\pi}{c^3} \frac{h\nu^3 d\nu}{e^{h\nu/k_B T} - 1}$$

You may want to use the geometric sum

$$S_n \equiv \sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$$

(8 marks)

7. Recover Rayleigh-Jeans law from part 4 by using Planck's blackbody distribution

(2 marks)

4 B question : Solution

- 1.

$$\begin{aligned}\psi(0) &= 0 \rightarrow B = 0 \\ \psi(L) &= A \sin(kL) = 0 \\ A \neq 0 \therefore kL &= n\pi \rightarrow k_n = \frac{n\pi}{L}\end{aligned}$$

(2 marks)

By normalising $\psi(x) = A \sin(kx)$

$$\int_0^L \psi^* \psi dx = 1 \rightarrow A^2 \int_0^L \sin^2(kx) dx = 1 \rightarrow A^2 = \frac{2}{L}$$

(2 marks)

2.

$$k_x = \frac{n_x \pi}{L}, \quad k_y = \frac{n_y \pi}{L}, \quad k_z = \frac{n_z \pi}{L}$$

$$\omega = ck, \quad k = \sqrt{k_x^2 + k_y^2 + k_z^2}$$

(1 mark)

3. In the large L approximation, ν becomes continuous and thus we can consider a spherical shell in the frequency space width $d\nu$, :

$$N(\nu) d\nu = 2 \times \frac{1}{8} \times \frac{4\pi\nu^2 d\nu}{(c/2L)^3} = \frac{8\pi L^3}{c^3} \nu^2 d\nu$$

(1 mark)

The factor of 2 is due to the two different polarisation of the wavevector
The factor of 1/8 arises as we're only consider the positive part of the sphere and hence the octant where $n_x, n_y, n_z > 0$

(1 mark)

4. The equipartition theorem states that each degree of freedom contributes an average energy of $\frac{k_B T}{2}$

(1 mark)

Hence the average thermal energy per unit volume in the interval is :

$$u(\nu) d\nu = 2 \times \frac{k_B T}{2} \times \frac{N(\nu) d\nu}{L^3} = \frac{8\pi k_B T}{c^3} \nu^2 d\nu$$

(1 mark)

5. Integrating over all energies is divergent and gives an infinite total energy per unit volume which is unphysical

Something very similar is required for the mark. Keywords include integration over all frequencies, infinite energy (1 mark)

6. Let $\beta = \frac{1}{k_B T}$. The normalisation factor $Z(\beta)$ is found by saying that the sum over all m should give you a probability of 1. Hence by using the geometric series formula with $r = e^{-\beta \epsilon}$ leads to

$$Z(\beta) = \sum_{(m=0)}^{\infty} e^{-m\beta\epsilon} = \frac{1}{1 - e^{-\beta\epsilon}}$$

(3 marks)

Recall that the expectation value $\langle x \rangle$ is given by $\langle x \rangle = \sum x f(x)$, and hence the average energy in this mode is then given by:

$$\frac{1}{Z(\beta)} \sum_{(m=0)}^{\infty} m\epsilon e^{-m\beta\epsilon} = -\frac{1}{Z(\beta)} \frac{\partial Z(\beta)}{\partial \beta} = \frac{\epsilon}{e^{\beta\epsilon} - 1} = \frac{h\nu}{e^{h\nu/k_B T} - 1}$$

(3 marks)

Hence the average thermal energy per unit volume in range ν to $\nu + d\nu$ is given by:

$$u(\nu) d\nu = \frac{h\nu}{e^{h\nu/k_B T} - 1} \frac{N(\nu) d\nu}{L^3} = \frac{8\pi}{c^3} \frac{h\nu^3 d\nu}{e^{h\nu/k_B T} - 1}$$

(2 marks)

7. By considering the low frequency limit, we can expand the Planck distribution

$$e^{h\nu/k_B T} \approx 1 + \frac{h\nu}{k_B T}$$

and return back to the expression in question 4

Taylor expansion must be explicitly shown (1 mark)

Correct evaluation for the expression (1 mark)

(2 marks)